# Homework 1 — changed numbers (extra practice)

Module 1 · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A pharmacist dissolved 300 milligrams of a new antibiotic drug into a volume of purified water and placed the solution in a refrigerator (4°C). At various time intervals, the pharmacist removed a small aliquot from the solution and measured the amount of drug contained in each aliquot. The following data were obtained.

| Time (hr) | Drug (mcg/mL) |
|---|---|
| 0.5 | 226.6 |
| 1 | 213.9 |
| 2 | 190.7 |
| 4 | 151.5 |
| 8 | 95.6 |
| 12 | 60.4 |
| 16 | 38.1 |

**1a** Module 1

Is the decomposition of this antibiotic a first-order or a zero-order process?

A. Zero order

B. First order

C. Neither: it needs a third plot

**1b** Module 1

What is the rate constant k for the decomposition of this antibiotic?

**1c** Module 1

What is the half-life t½?

**1d** Module 1

What was the initial volume of water used to prepare the original solution?

**1e** Module 1

How much time is required for the concentration of drug to decrease by 75%?

2. Below is the decrease in the amount of drug as a function of time.

| Time (hr) | Drug A (mg) |
|---|---|
| 0.5 | 508 |
| 1 | 496 |
| 2 | 472 |
| 3 | 448 |
| 6 | 376 |
| 9 | 304 |
| 12 | 232 |

| Time (hr) | Drug A (mg) |
|---|---|
| 16 | 136 |

**2a** Module 1

Does the decrease in the amount of drug A appear to be a zero-order or a first-order process?

A. Cannot tell without logarithms

B. First order

C. Zero order

**2b** Module 1

What is the rate constant k?

**2c** Module 1

What is the half-life t½?

3. A solution of a drug was freshly prepared at a concentration of 280 mg/mL. After 30 days at 25°C, the drug concentration in the solution was 84 mg/mL.

**3a** Module 1

Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

**3b** Module 1

Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?

# Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

## 1a

Worked by the drill

**Answer:** B. First order

Over each 4-hour step from 4 to 16 hr the concentration falls by the same fraction (1.58, 1.58, 1.59), so ln C against time is a straight line. A constant fraction lost per equal time is first order, because the rate is proportional to the amount remaining.

Her homework wording with different numbers, so this is not the graded answer.

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## 1b

Worked by the drill

**Answer:** 0.115 $hr^{-1}$

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 300 milligrams | *not needed: k comes from the slope, not the amount* |
| **temperature** | 4°C | *not needed* |
| **$C_1$** | 213.9 mcg/mL at 1 hr | $\ln(C_1)$, the first point |
| **$C_2$** | 38.1 mcg/mL at 16 hr | $\ln(C_2)$, the second point |
| **$t_2 - t_1$** | 16 hr − 1 hr | the denominator, 15 hr |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(C_1) - \ln(C_2)}{t_2 - t_1}, \text{ with } t_1 = 1 \text{ hr and } t_2 = 16 \text{ hr}$$

For first order, $\ln C = \ln C_0 - kt$, so ln C against t is a straight line with slope −k, where k is the first-order rate constant. Two points far apart give the slope with the least error; her key uses the 1 hr and 16 hr points, 15 hours apart.

ALGEBRA

$$k = \frac{\ln(213.9) - \ln(38.1)}{15 \text{ hr}} = \frac{1.7253}{15 \text{ hr}} = 0.1150 \text{ hr}^{-1}$$

$\ln(213.9) - \ln(38.1) = \ln(\frac{213.9}{38.1}) = 1.7253$: the mcg/mL cancel inside the logarithm, leaving a pure number. Dividing by the 15 hr between the points leaves reciprocal hours, so k = 0.1150 $hr^{-1}$, the fraction of the drug lost per hour.

> **DOES THE NUMBER MAKE SENSE?**
>
> Between 2 and 8 hr the level falls from 190.7 to 95.6, about half, so the half-life is near 6 hr and k near $\frac{0.693}{6}$, a little over 0.115 $hr^{-1}$; the two-point slope gives 0.1150 $hr^{-1}$.

Her homework wording with different numbers, so this is not the graded answer.

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## 1c Worked by the drill

**Answer:** 6.03 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 300 milligrams | *not needed: a first-order half-life does not depend on the amount* |
| **table** | 213.9 mcg/mL at 1 hr and 38.1 mcg/mL at 16 hr | gave k = 0.1150 $hr^{-1}$ in part b, the denominator |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693}{k} = \frac{0.693}{0.1150\ hr^{-1}} = 6.03\ hr$$

The half-life t½ is the time for C to fall to half, so $e^{-k\,t½} = 0.5$ and $k\,t½ = \ln 2 = 0.693$, giving $t½ = \frac{0.693}{k}$. With k = 0.1150 $hr^{-1}$ from part b, the reciprocal hours in the denominator invert to hours: 6.03 hr for every halving.

> **DOES THE NUMBER MAKE SENSE?**
>
> From 2 hr (190.7) to 8 hr (95.6) the level falls to about half, so the half-life is near 6 hr; $\frac{0.693}{0.1150}$ = 6.03 hr agrees.

Her homework wording with different numbers, so this is not the graded answer.

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## 1d Worked by the drill

**Answer:** 1250 mL

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 300 milligrams | written 300,000 mcg, the numerator |
| **$C_1$** | 213.9 mcg/mL at 1 hr | brought back 1 hr to $C_0$ = 240.0 mcg/mL, the denominator |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$\ln C_0 = \ln C + kt = \ln(213.9) + (0.1150)(1) \rightarrow C_0 = 240.0\ mcg/mL$$

The first-order line $\ln C = \ln C_0 - kt$ rearranges to $\ln C_0 = \ln C + kt$, with $C_0$ the concentration the moment the drug dissolved. The 1 hr point, 213.9 mcg/mL, has declined for 1 hour at k = 0.1150 $hr^{-1}$; adding kt = 0.1150 undoes that decline: $C_0$ = 240.0 mcg/mL.

UNIT

$$300\ mg \times 1000\ mcg/mg = 300{,}000\ mcg$$

The 300 mg dissolved is the amount at time zero. The concentration is in mcg/mL, so the amount is put in micrograms, 300 × 1000 mcg/mg = 300,000 mcg; then mcg over mcg/mL cancels the micrograms and leaves millilitres, the unit asked for the volume.

ALGEBRA

$$V = \frac{300{,}000 \text{ mcg}}{240.0 \text{ mcg/mL}} = 1250 \text{ mL}$$

Concentration is amount over volume, $C_0 = \frac{D_0}{V}$, so $V = \frac{D_0}{C_0}$: the whole 300,000 mcg spread through the water at 240.0 mcg/mL. The micrograms cancel and millilitres are left, 1250 mL, the volume of purified water the pharmacist used.

> **DOES THE NUMBER MAKE SENSE?**
> $C_0$ must be a little above the 0.5 hr value of 226.6 mcg/mL; it is 240.0. Then 300,000 mcg at 240.0 mcg/mL is $\frac{300{,}000}{240.0}$, a bit over 1000 mL: 1250 mL.

Her homework wording with different numbers, so this is not the graded answer.

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## 1e

Worked by the drill

**Answer:** 12.05 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 300 milligrams | *not needed: the time to a fraction does not depend on the amount* |
| **fraction** | decrease by 75% | leaves 25%, so $C_0$ over C is 4, two half-lives |
| **table** | 213.9 mcg/mL at 1 hr and 38.1 mcg/mL at 16 hr | gave $k = 0.1150$ hr$^{-1}$ in part b |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$t = \frac{\ln(C_0/C)}{k} = \frac{\ln(1/0.25)}{0.1150 \text{ hr}^{-1}}$$

The first-order line $\ln C = \ln C_0 - kt$ solved for the time is $t = \frac{\ln(C_0/C)}{k}$. A 75% decrease leaves 25% of the starting concentration, so $C_0$ over C = $\frac{1}{0.25} = 4$ whatever the starting value, and $k = 0.1150$ hr$^{-1}$ is the rate constant from part b.

ALGEBRA

$$t = \frac{1.3863}{0.1150 \text{ hr}^{-1}} = 12.05 \text{ hr} = 2 \times 6.025 \text{ hr}$$

$\ln 4 = 1.3863$, a pure number, so dividing by k in hr$^{-1}$ leaves hours: $\frac{1.3863}{0.1150 \text{ hr}^{-1}} = 12.05$ hr. Falling to 25% is two halvings, 100% to 50% to 25%, so the time is exactly $2 \times t_{½} = 2 \times 6.025$ hr, which checks the answer against part c.

> **DOES THE NUMBER MAKE SENSE?**

> Losing 75% leaves a quarter, two halvings, so the time is two half-lives: 2 × 6.025 hr = 12.05 hr, more than the one half-life of 6.025 hr.

Her homework wording with different numbers, so this is not the graded answer.

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## 2a Worked by the drill

**Answer:** C. Zero order

Each 3-hour step from 3 to 12 hr loses the same 72 mg, so the amount against time is a straight line on ordinary axes. A constant amount lost per equal time is zero order, because the rate does not depend on how much drug is left.

Her homework wording with different numbers, so this is not the graded answer.

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## 2b Worked by the drill

**Answer:** 24 mg/hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **A1** | 496 mg at 1 hr | the first amount |
| **A2** | 136 mg at 16 hr | the second amount |
| **$t_2 - t_1$** | 16 hr − 1 hr | the denominator, 15 hr |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$k_0 = \frac{A1 - A2}{t_2 - t_1}$$

For zero order the amount falls by the same quantity each hour, $A = A0 - k_0\, t$, so $k_0$, the zero-order rate constant, is the constant amount lost per unit time, the slope of amount against time: the difference between two amounts over the time between them, in mg/hr.

ALGEBRA

$$k_0 = \frac{496\ \text{mg} - 136\ \text{mg}}{16\ \text{hr} - 1\ \text{hr}} = \frac{360\ \text{mg}}{15\ \text{hr}} = 24\ \text{mg/hr}$$

496 mg at 1 hr minus 136 mg at 16 hr is 360 mg lost over 16 − 1 = 15 hr; $\frac{360\ \text{mg}}{15\ \text{hr}} = 24$ mg/hr. Milligrams over hours is an amount per unit time, the unit of a zero-order rate constant, unlike the $\text{hr}^{-1}$ of first order.

> **DOES THE NUMBER MAKE SENSE?**
> From 3 hr (448 mg) to 6 hr (376 mg) the drop is the same as from 9 to 12 hr, 304 to 232 mg, a constant loss; 496 − 136 = 360 mg over 15 hr is 24 mg/hr.

Her homework wording with different numbers, so this is not the graded answer.

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## 2c Worked by the drill

**Answer:** 10.83 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **A at 0.5 hr** | 508 mg | brought back 0.5 hr to A0 = 520 mg, the numerator |
| **table** | 496 mg at 1 hr and 136 mg at 16 hr | gave $k_0$ = 24 mg/hr in part b, in the denominator $2k_0$ |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$A0 = A + k_0\,t = 508\text{ mg} + (24\text{ mg/hr})(0.5\text{ hr}) = 520\text{ mg}$$

The zero-order line $A = A0 - k_0\,t$ rearranges to $A0 = A + k_0\,t$, with A0 the amount at time zero. The first entry, 508 mg at 0.5 hr, has already lost $k_0\,t = (24\text{ mg/hr})(0.5\text{ hr}) = 12$ mg; adding it back gives A0 = 520 mg for the half-life.

ALGEBRA

$$t½ = \frac{A0}{2k_0} = \frac{520\text{ mg}}{2(24\text{ mg/hr})} = 10.83\text{ hr}$$

Half of A0 is 260 mg, and at 24 mg/hr that loss takes $\frac{260\text{ mg}}{24\text{ mg/hr}} = 10.83$ hr, which is $t½ = \frac{A0}{2k_0}$. Unlike first order, this half-life grows with the starting amount, because a constant loss takes longer to remove half of a larger amount.

> **DOES THE NUMBER MAKE SENSE?**
> Half of A0 = 520 mg lies between the 304 mg at 9 hr and the 232 mg at 12 hr, so t½ is between 9 and 12 hr: 10.83 hr.

Her homework wording with different numbers, so this is not the graded answer.

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## 3a

Worked by the drill

**Answer:** 17.27 days

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$C_0$** | 280 mg/mL | the numerator of the ratio inside the logarithm |
| **t** | 30 days | the denominator of k |
| **temperature** | 25°C | *not needed* |
| **C** | 84 mg/mL | the denominator of the ratio inside the logarithm |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(280/84)}{30\text{ days}} = \frac{1.2040}{30\text{ days}} = 0.04013\text{ day}^{-1}$$

First order: $\ln C = \ln C_0 - kt$, so $k = \frac{\ln(C_0/C)}{t}$, the log of the ratio of starting to final concentration over the time between them. $\frac{280}{84} = 3.333$, a pure number because the mg/mL cancel, and ln 3.333 = 1.2040; the time is in days, so k is in $\text{day}^{-1}$.

ALGEBRA

$$t½ = \frac{0.693}{0.04013\ \text{day}^{-1}} = 17.27\ \text{days}$$

$t½ = \frac{0.693}{k}$ because $k\ t½ = \ln 2$ when C falls to half; the reciprocal days in k invert to days: $\frac{0.693}{0.04013\ \text{day}^{-1}} = 17.27$ days. A first-order half-life does not depend on the starting concentration: every halving takes the same 17.27 days.

**DOES THE NUMBER MAKE SENSE?**

280 to 84 mg/mL is a fall to under a third in 30 days, more than one halving but less than two, so t½ is between 15 and 30 days: 17.27 days.

Her homework wording with different numbers, so this is not the graded answer.

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## 3b

Worked by the drill

**Answer:** 21.43 days

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$C_0$** | 280 mg/mL | in $k_0$ and the numerator of $\frac{C_0}{2k_0}$ |
| **t** | 30 days | the denominator of $k_0$ |
| **temperature** | 25°C | *not needed* |
| **C** | 84 mg/mL | subtracted from $C_0$ in $k_0$ |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$k_0 = \frac{280 - 84\ \text{mg/mL}}{30\ \text{days}} = 6.5333\ \text{(mg/mL)/day}$$

Zero order: $C = C_0 - k_0\,t$, so $k_0 = \frac{C_0 - C}{t}$, the constant loss per unit time. 280 − 84 = 196 mg/mL was lost over 30 days, so $k_0 = 6.5333$ (mg/mL)/day, a concentration per unit time rather than the $\text{day}^{-1}$ of a first-order k.

ALGEBRA

$$t½ = \frac{C_0}{2k_0} = \frac{280\ \text{mg/mL}}{2(6.5333\ \text{(mg/mL)/day})} = 21.43\ \text{days}$$

Half of the original 280 mg/mL is 140 mg/mL; at a constant 6.5333 (mg/mL)/day that loss takes $\frac{140\ \text{mg/mL}}{6.5333\ \text{(mg/mL)/day}} = 21.43$ days, which is $t½ = \frac{C_0}{2k_0}$. The mg/mL cancel and days are left; the zero-order half-life uses the starting concentration, so it differs from the first-order 17.27 days of part a.

**DOES THE NUMBER MAKE SENSE?**

Zero order loses the same amount each day. In 30 days more than half but less than three-quarters of the 280 mg/mL was lost, so halving takes between 20 and 30 days: 21.43 days.

Her homework wording with different numbers, so this is not the graded answer.

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